

Notes on MATH 375: Probability Theory

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Lecture 2

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1 Lecture 2: Counting and Probability

1.1 Permutations with repeated objects

If n objects include groups of identical objects, arranging all n positions gives $n!$ orderings, but swapping identical objects does not create a new arrangement. Divide by the factorial of each group size.

Example: SUCCESS. The word SUCCESS has seven letters: three S's, two C's, one U, and one E. The number of distinct arrangements is

$$\frac{7!}{3! 2! 1! 1!} = 420.$$

1.2 Combinations

A **combination** selects objects without regard to their order. The number of ways to choose r objects from n distinct objects is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Here n is the number available and r is the number chosen.

For example,

$$\binom{5}{3} = \frac{5!}{3! 2!} = 10.$$

Choosing the three objects to include is equivalent to choosing the two objects to leave out, so

$$\binom{n}{r} = \binom{n}{n-r}.$$

Example: a required letter. How many ways can we select eight letters from the English alphabet if A must be included? A is already selected, so choose the other seven letters from the remaining 25:

$$\boxed{\binom{25}{7}}.$$

Example: exactly three zeros. A bit string of length 10 with exactly three zeros is determined by the three positions containing zeros:

$$\binom{10}{3} = 120.$$

Example: at least three zeros. Start with all bit strings of length 10. Exclude those with zero, one, or two zeros:

$$\begin{aligned} \#(\text{at least three zeros}) &= 2^{10} - \left[\binom{10}{0} + \binom{10}{1} + \binom{10}{2} \right] \\ &= 1024 - (1 + 10 + 45) \\ &= \boxed{968}. \end{aligned}$$

This is the **complement method**: count everything, then subtract the cases that fail the condition.

1.3 Sample spaces and events

The **sample space** S is the set of all possible outcomes. An **event** E is a subset of S : the outcomes that satisfy a particular condition.

When all outcomes in a finite sample space are equally likely,

$$P(E) = \frac{|E|}{|S|} = \frac{\text{number of favorable outcomes}}{\text{number of possible outcomes}}.$$

The denominator must count the outcomes in the sample space, not just the outcomes in the event.

Example: rolling a die. For a fair six-sided die,

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Let E be the event “the result is odd.” Then

$$E = \{1, 3, 5\}, \quad P(E) = \frac{|E|}{|S|} = \frac{3}{6} = \frac{1}{2}.$$

1.4 Basic probability rules

For every event E ,

$$0 \leq P(E) \leq 1.$$

The probability of the entire sample space is

$$P(S) = 1.$$

If E and F are **mutually exclusive**, they cannot occur together: $E \cap F = \emptyset$. In that case,

$$P(E \cup F) = P(E) + P(F).$$

1.5 Union, intersection, and complement

$E \cup F$ means **E or F or both** occur.

$E \cap F$ means **E and F** occur together.

The complement E^c means **E does not occur**. Since E and E^c are mutually exclusive and together make up S ,

$$P(E) + P(E^c) = 1.$$

Therefore, the **complement rule** is

$$\boxed{P(E^c) = 1 - P(E)}.$$

For example, if $P(E) = \frac{1}{6}$, then

$$P(E^c) = 1 - \frac{1}{6} = \frac{5}{6}.$$